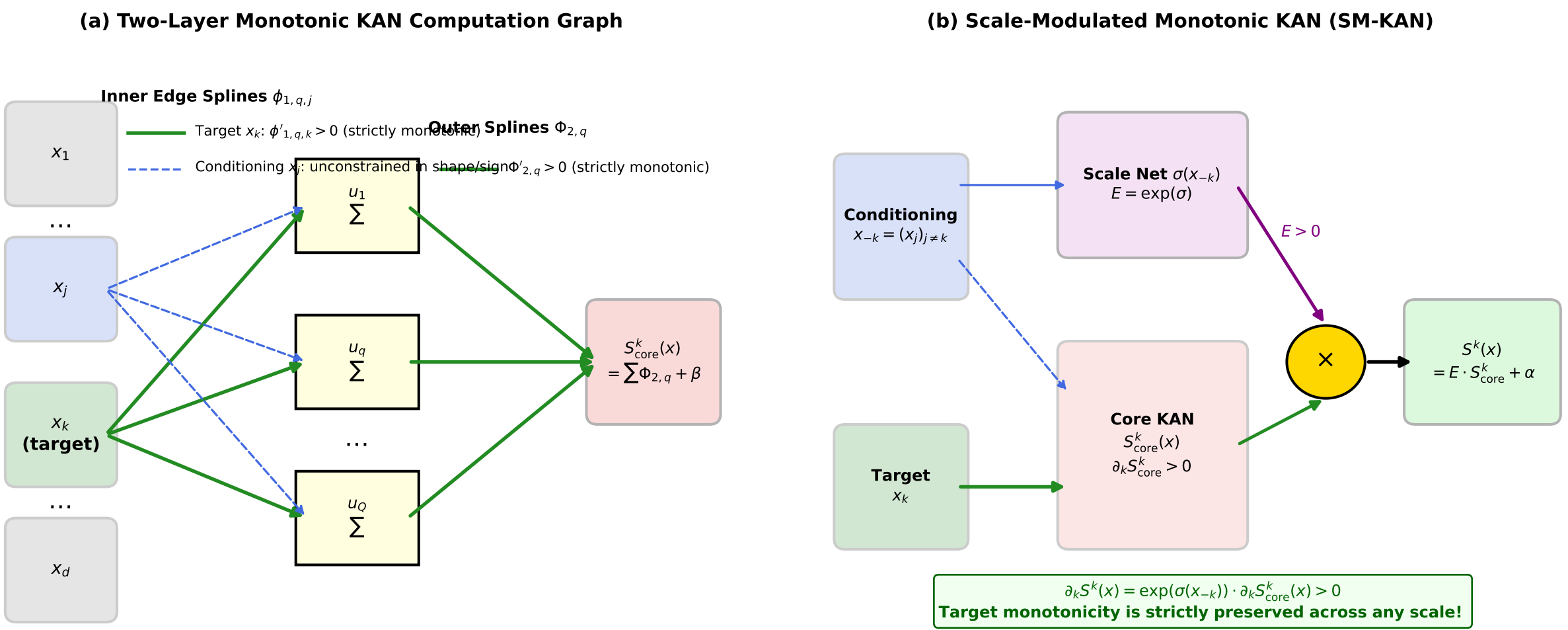
(a) Two-Layer Monotonic KAN Computation Graph
Inner Edge Splines $\phi_{1,q,j}$
Target x_k: $\phi'_{1,q,k} > 0$ (strictly monotonic)
Conditioning x_j: unconstrained in shape/sign
Outer Splines $\Phi_{2,q}$
$\Phi'_{2,q} > 0$ (strictly monotonic)
x_1
x_j
x_k (target)
x_d
u_1
u_q
u_Q
$S^k_{\text{core}}(x) = \sum \Phi_{2,q} + \beta$
(b) Scale-Modulated Monotonic KAN (SM-KAN)
Conditioning $x_{-k} = (x_j)_{j \neq k}$
Scale Net $\sigma(x_{-k})$ $E = \exp(\sigma)$
$E > 0$
Target x_k
Core KAN $S^k_{\text{core}}(x)$ $\partial_k S^k_{\text{core}} > 0$
$S^k(x) = E \cdot S^k_{\text{core}} + \alpha$
$\partial_k S^k(x) = \exp(\sigma(x_{-k})) \cdot \partial_k S^k_{\text{core}}(x) > 0$
Target monotonicity is strictly preserved across any scale!